

GRAIN AND TILLER: LOSS-INFORMED RATIONAL TRANSFORMER UPDATES

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ABSTRACT

1 INTRODUCTION

2 RELATED WORK

3 METHOD

We introduce two coupled components. GRAIN is a grouped rational feed-forward activation whose normalization exposes a compact, differentiable response geometry. TILLER is an optimizer that uses this geometry and downstream loss derivatives to coordinate structured matrix directions. Its central operation is a global transaction over layer-group coordinates: it redistributes a fixed update budget among signed coefficients instead of changing the global learning-rate budget.

Let a Transformer have L feed-forward blocks, G rational groups per block, model width d , and expanded rational width $H_r = Gw$. We index the $C = LG$ layer-group coordinates by $i = (\ell, g)$. All inner products and norms below are Frobenius inner products and norms after the relevant matrix has been viewed in these coordinates.

3.1 GRAIN: GROUPWISE RATIONAL ACTIVATION WITH INTERNAL NORMALIZATION

Gated feed-forward networks use multiple projections to create a multiplicative feature map (Dauphin et al., 2017; Shazeer, 2020). GRAIN instead learns the feature map itself. For an input $x \in \mathbb{R}^d$, one block is

$$z = Ax, \quad h = \Phi(z), \quad y = Bh, \quad (1)$$

where $A \in \mathbb{R}^{H_r \times d}$ and $B \in \mathbb{R}^{d \times H_r}$. We partition z into G contiguous groups $z_g \in \mathbb{R}^w$ and define

$$\rho_g = \sqrt{w^{-1} \|z_g\|_F^2 + \epsilon_{\text{act}}}, \quad u_g = z_g / \rho_g, \quad \epsilon_{\text{act}} = 10^{-6}. \quad (2)$$

Each group has a learned P5/Q4 rational response

$$P_g(u) = \sum_{k=0}^5 a_{gk} u^k, \quad (3)$$

$$Q_g(u) = 1 + \sum_{k=1}^4 |b_{gk}| |u|^k, \quad (4)$$

$$\Phi_g(z_g) = \rho_g \phi_g(u_g), \quad \phi_g(u) = P_g(u) / Q_g(u), \quad (5)$$

with elementwise powers. This combines the flexibility of learned rational activations (Molina et al., 2020; Boullé et al., 2020) with an internal RMS coordinate system (Zhang & Sennrich, 2019). Since $Q_g(u) \geq 1$, the learned response has no real denominator poles.

For a SwiGLU intermediate width H , we set $H_r = 3H/2$. The two dense matrices in equation 1 then contain $2dH_r = 3dH$ weights, the same dense projection count as the three SwiGLU matrices. The activation adds only $10G$ learned rational scalars per block. The group width is capped at 256, keeping both the activation kernel and its response summaries local and linear in H_r .

The normalized map has an especially useful Jacobian. Writing $f = \phi_g(u_g)$, $d_g = \phi'_g(u_g)$, and $r_g = f - u_g \odot d_g$, differentiation gives

$$J_g \equiv \frac{\partial \Phi_g}{\partial z_g} = \text{Diag}(d_g) + \frac{1}{w} r_g u_g^\top, \quad J_g^\top v = d_g \odot v + u_g \text{ mean}(r_g \odot v). \quad (6)$$

Thus a group response and its exact pullback require a diagonal operation and one rank-one correction, without materializing a $w \times w$ Jacobian.

3.2 RESPONSE COORDINATES AND STRUCTURED DIRECTIONS

For each structured matrix role, TILLER starts from the Nesterov–Muon direction p_i : momentum is formed first, followed by the five-step Newton–Schulz polar map and Muon’s role-dependent RMS adjustment (Jordan et al., 2024; Bernstein & Newhouse, 2024; Liu et al., 2025). In parallel, it maintains factorized row and column second moments, in the spirit of memory-efficient factored adaptation (Shazeer & Stern, 2018). With gradient g_i and Nesterov momentum m_i ,

$$R_t = \beta R_{t-1} + (1 - \beta) \text{rowsum}(g_i^2), \quad (7)$$

$$C_t = \beta C_{t-1} + (1 - \beta) \text{colsum}(g_i^2), \quad (8)$$

$$V_{t,jk} = \frac{R_{t,j} C_{t,k}}{\sum_s R_{t,s}} \frac{1}{1 - \beta^t}, \quad a_i = \frac{m_i}{\sqrt{V_t} + \epsilon_{\text{opt}}}. \quad (9)$$

The implementation uses $\beta = 0.95$ and $\epsilon_{\text{opt}} = 10^{-8}$.

Two bounded summaries determine how much of this adaptive information enters the update. The participation $\pi_i \in [0, 1]$ measures the effective rank of the live group response, using the Jacobian spectrum for the incoming role and the feature-energy spectrum for the outgoing role. The congruence $\kappa_i \in [0, 1]$ is a normalized kernel alignment between the live response and the response at initialization. Both are weighted by downstream cotangent energy. We allocate

$$e_i = \pi_i (1 - \kappa_i^2) \quad (10)$$

of the direction energy to the adaptive tangent. Let

$$a_i^\perp = a_i - \frac{\langle a_i, p_i \rangle}{\|p_i\|_F^2} p_i, \quad s_i = \text{sign} \langle g_i, a_i^\perp \rangle, \quad (11)$$

with $s_i = 1$ for a zero inner product. The structured chord is

$$q_i = \sqrt{1 - e_i} p_i + s_i \sqrt{e_i} \frac{\|p_i\|_F}{\|a_i^\perp\|_F} a_i^\perp. \quad (12)$$

Because $a_i^\perp \perp p_i$, the chord preserves the parent norm exactly. A nonfinite or degenerate chord, or one that fails positive first-order descent for either parent or candidate, reverts locally to p_i .

The intuition in equations 10–12 is geometric. A concentrated response provides little support for a factored tangent, while a diffuse response can support one. A response that still matches its initialization keeps the Muon direction; learned departure opens an angular budget. The current gradient chooses the tangent orientation, so the construction does not impose a nonnegative coefficient convention.

3.3 EXACT LOSS-COORDINATE MEASUREMENTS

Every eighth optimizer step, TILLER records $K = 32$ fixed functional probes distributed across the current batch. For one probe in coordinate $i = (\ell, g)$, let x be the block input, $z = Ax$, $h = \Phi(z)$, $c = \partial \mathcal{L} / \partial y$, and let $(\Delta A_i, \Delta B_i)$ be the two structured directions after their Muon adjustments. The ledger entry is

$$S_{k,i} = \langle x \Delta A_i^\top, J_i^\top B^\top c \rangle + \langle h_i, c \Delta B_i \rangle. \quad (13)$$

This is the exact first-order directional derivative of the downstream loss along the paired group direction. Equation 6 evaluates it without a residual-sized tangent image. Applying the same contraction to the decoupled decay direction yields a probe-wise decay action.

On intermediate steps, the two matrix roles provide exact current-gradient scores

$$h_{t,i} = \langle g_{t,i}^A, q_{t,i}^A \rangle + \langle g_{t,i}^B, q_{t,i}^B \rangle. \quad (14)$$

Let ν be the RMS row norm from the latest functional measurement. For $\|h_t\|_2 > 0$, the surrogate row is $\bar{s}_t = (\nu/\|h_t\|_2)h_t$, repeated K times. It therefore preserves the calibrated total score energy while incorporating the current gradient on every optimizer step.

3.4 A ROBUST FREQUENT-DIRECTIONS LOSS LEDGER

The score matrix has $C = LG$ columns, but TILLER never forms a $C \times C$ covariance. It retains a rank- r factor B_{t-1} with $r = 64$ and forms

$$A_t = \begin{bmatrix} \sqrt{\beta} B_{t-1} \\ \sqrt{1-\beta} S_t \end{bmatrix}, \quad (15)$$

whose live row count is at most $r + K = 96$. Following Frequent Directions and its robust isotropic completion (Ghashami et al., 2016; Luo et al., 2019), TILLER diagonalizes only the row Gram matrix $A_t A_t^\top$. If Δ is the largest discarded eigenvalue, it shrinks every retained eigenvalue by Δ and adds $\Delta/(2K)$ to an isotropic tail τ_t . The represented metric is

$$\widehat{M}_t = \frac{1}{K} B_t^\top B_t + \tau_t I, \quad \tau_t = \beta \tau_{t-1} + \frac{\Delta}{2K}. \quad (16)$$

The midpoint tail treats the unresolved interval symmetrically: relative to the uncompressed live metric, its worst-direction spectral error is at most $\Delta/(2K)$ (Appendix D). A decay cross term is advanced with the same β recurrence.

3.5 SIGNED EQUAL-BUDGET TRANSACTION

Let $c \in \mathbb{R}^C$ scale the paired structured directions and define $w_i = \|q_i^A\|_F^2 + \|q_i^B\|_F^2$. The parent Muon transaction is $c = \mathbf{1}$ with budget $B_0 = \sum_i w_i$. With current exact gradient score h , ledger factor S , diagonal completion d , decay cross r_d , and scheduled step size η , TILLER solves the projected problem

$$\min_c \quad F(c) = -\eta h^\top c + \frac{\eta^2}{2} \left[\frac{1}{K} \|Sc\|_2^2 + d^\top (c \odot c) + 2r_d^\top c \right], \quad (17)$$

$$\text{s.t.} \quad \sum_i w_i c_i^2 = B_0. \quad (18)$$

The coefficients are signed: the transaction may amplify, attenuate, or reverse a coordinate while preserving the global weighted budget.

After whitening $y = W^{1/2}c$, the stationarity equation is

$$(M + \lambda I)y = b, \quad M = W^{-1/2} (S^\top S/K + \text{Diag}(d)) W^{-1/2}, \quad b = W^{-1/2} (h/\eta - r_d). \quad (19)$$

We build a fully reorthogonalized 32-vector Lanczos basis using only $v \mapsto d \odot v + S^\top(Sv)/K$ (Lanczos, 1950), diagonalize the resulting at-most 32×32 tridiagonal, and find λ by fixed bisection. The proposed coefficients are reevaluated in the full low-rank-plus-diagonal objective. They are accepted only if the budget residual is at most 10^{-8} , $F(c) < F(\mathbf{1})$, all quantities are finite, and current-gradient and Nesterov contractions remain positive for both matrix roles in every layer. Otherwise the parent vector $\mathbf{1}$ is used.

3.6 ATTENTION, REMAINING PARAMETERS, AND DISTRIBUTED REALIZATION

The attention QKV and output matrices use the same response-informed, norm-preserving tangent chord, applied to head-blocked Muon directions, but do not enter the LG coefficient transaction. The learned rational coefficients, embeddings, normalization parameters, and other remaining tensors use AdamW (Loshchilov & Hutter, 2019).

The global algebra is owner-free. A shard stores only its parameter fragments, factorized moments, score columns, and coefficient segment. Tensor fragments contribute additive inner products; coordinate shards contribute additive $K \times K$ row Grams, layer certificates, and Krylov dot products. Collective reductions combine these compact quantities, and each shard reconstructs its own update from the shared scalar solution. This is compatible with tensor, fully sharded, data, and pipeline decompositions (Zhao et al., 2023; Shi et al., 2023); a replicated data-parallel layout is the special case in which every rank already holds all C coordinates. Neither a complete layer nor a parameter-sized selected update is assigned to an owner.

The method-specific persistent floating-point state, excluding the baseline Muon momentum and AdamW moments, is

$$112LG + 2L + 38 + 2LH_r + 2LGd + 6Ld = O(LH_r + LGd + 64LG). \quad (20)$$

The ledger itself uses $66LG + 4$ elements, and its maximum live factor uses $96LG$ elements. The largest temporal eigendecomposition is 96×96 ; the largest transaction eigendecomposition is 32×32 . All ledger work is linear in LG up to these fixed row bounds. Activation evaluation is $O(LNH_r)$ for N activation positions, while optimizer state is independent of N .

4 EXPERIMENTS

5 DISCUSSION

6 CONCLUSION

AI USE STATEMENT

Generative AI tools assisted with code implementation, mathematical cross-checking, literature retrieval, and manuscript drafting. All resulting mathematical statements, citations, code paths, and experimental records were checked against the implementation, executable artifacts, and primary sources. The authors take responsibility for the final content of this work.

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A EXACT GRAIN CALCULUS

Proposition 1 (Pole freedom and Jacobian structure). *For every finite group input, $Q_g(u) \geq 1$. Wherever the absolute-value terms are differentiable, the Jacobian of equations 2 and 5 is equation 6. At zero, the implementation uses the subgradient convention $\text{sign}(0) = 0$.*

Proof. Every term $|b_{gk}| |u|^k$ is nonnegative, hence $Q_g(u) \geq 1$. For the Jacobian, suppress the group index and write $\rho = (w^{-1}z^\top z + \epsilon_{\text{act}})^{1/2}$ and $u = z/\rho$. Then

$$\nabla_z \rho = u/w, \quad \frac{\partial u}{\partial z} = \rho^{-1} (I - uu^\top/w). \quad (21)$$

Applying the product and chain rules to $\Phi(z) = \rho\phi(u)$ gives

$$J = \phi(u)u^\top/w + \text{Diag}(\phi'(u)) (I - uu^\top/w) \quad (22)$$

$$= \text{Diag}(d) + (\phi(u) - u \odot d) u^\top/w. \quad (23)$$

Transposition and multiplication by v yield the pullback in equation 6. The derivation retains ϵ_{act} throughout. $\square$

For reproducibility, all groups begin from the same SiLU fit. The numerator coefficients, in ascending powers, are

$$a^{\text{init}} = \text{vec} \begin{pmatrix} 2.4915714620520875 \times 10^{-5} & 0.5000668663485823 \\ 0.2499441908995771 & 0.0526200910151016 \\ 0.005525765640115002 & 0.00024199321178747251 \end{pmatrix}, \quad (24)$$

and the four denominator coefficients are

$$b^{\text{init}} = \text{vec} \begin{pmatrix} 0.00022554017910995938 & 0.10511420350691994 \\ 2.8656992394109812 \times 10^{-5} & 0.0004816962110562025 \end{pmatrix}. \quad (25)$$

All ten coefficients then learn independently in every group.

B RESPONSE PARTICIPATION AND THE TANGENT CHORD

For a nonzero Jacobian with singular values $\{\sigma_j\}_{j=1}^w$, the incoming participation is

$$\pi_{\text{in}} = \frac{(\sum_j \sigma_j^2)^2}{w \sum_j \sigma_j^4}. \quad (26)$$

For a nonzero normalized feature f , the outgoing participation is

$$\pi_{\text{out}} = \frac{\|f\|_2^4}{w \|f\|_4^4}. \quad (27)$$

Probe values are averaged with squared downstream pullback or cotangent weights. Congruence is the corresponding normalized Frobenius alignment between live and initialization-time Jacobian Gram kernels for the incoming role, and between live and initialization-time feature kernels for the outgoing role.

Lemma 1 (Participation bounds). *For a nonzero response, both quantities in equations 26 and 27 lie in $[1/w, 1]$.*

Proof. For nonnegative v_j , Cauchy–Schwarz gives $(\sum_j v_j)^2 \leq w \sum_j v_j^2$, while $(\sum_j v_j)^2 \geq \sum_j v_j^2$. Apply this first to $v_j = \sigma_j^2$ and then to $v_j = f_j^2$. $\square$

Proposition 2 (Local budget and descent certificate). *Assume $p \neq 0$ and $a^\perp \neq 0$. The chord in equation 12 satisfies $\|q\|_F = \|p\|_F$. Whenever the implementation selects this chord rather than its fallback, it also has $\langle g, p \rangle > 0$ and $\langle g, q \rangle > 0$.*

Proof. By construction, $\langle p, a^\perp \rangle = 0$. Since $e \in [0, 1]$,

$$\|q\|_F^2 = (1 - e) \|p\|_F^2 + e \|p\|_F^2 = \|p\|_F^2. \quad (28)$$

The two strict inequalities are explicit predicates of the local acceptance test; failure selects p . $\square$

C LOSS-LEDGER IDENTITIES

Lemma 2 (Trace-matched intermediate row). *Let $\nu^2 = \|S_{\text{ref}}\|_F^2/K$ and $h \neq 0$. For $S = \mathbf{1}_K(\nu h / \|h\|_2)^\top$,*

$$\frac{1}{K} \|S\|_F^2 = \nu^2. \quad (29)$$

Proof. All K rows have squared norm $\nu^2 \|h\|_2^2 / \|h\|_2^2 = \nu^2$. $\square$

The identity matches total score energy, the exact current-gradient vector supplies the direction, and functional refreshes supply the full score covariance.

D ROBUST FREQUENT-DIRECTIONS MIDPOINT COMPLETION

Proposition 3 (PSD and spectral error). *Let*

$$M = A^\top A/K + \beta\tau_{t-1}I \quad (30)$$

be the live metric before compression. Retain the largest r eigenvalues of $AA^\top$, subtract the largest discarded eigenvalue Δ , and form $\widehat{M}$ as in equation 16. Then

$$\widehat{M} \succeq 0, \quad \|\widehat{M} - M\|_2 \leq \frac{\Delta}{2K}. \quad (31)$$

When no row is discarded, $\Delta = 0$.

Proof. The retained eigenvalues after shrinkage are nonnegative, and $\tau_t = \beta\tau_{t-1} + \Delta/(2K) \geq 0$, proving positive semidefiniteness. On a retained right-singular direction, compression changes the eigenvalue by exactly $-\Delta/(2K)$. On a discarded direction with live eigenvalue $\lambda \in [0, \Delta]$, the error is $(\Delta/2 - \lambda)/K \in [-\Delta/(2K), \Delta/(2K)]$. On a null direction it is $\Delta/(2K)$. The operator norm is the largest absolute eigenvalue error, which proves equation 31. $\square$

The scalar $\Delta/(2K)$ is the midpoint of the unresolved interval $[0, \Delta/K]$ and therefore minimizes the worst absolute scalar-completion error over that interval.

E TRANSACTION DERIVATION AND ACCEPTANCE GUARANTEES

Let $W = \text{Diag}(w)$ and set $y = W^{1/2}c$. Substituting $c = W^{-1/2}y$ into equations 17 and 18 gives, up to a constant positive factor,

$$\min_{\|y\|_2=B_0} \frac{1}{2}y^\top My - b^\top y, \quad (32)$$

with M and b from equation 19. Its Lagrangian stationarity condition is $(M + \lambda I)y = b$. In the Lanczos subspace $\mathcal{K}_{32}(M, b)$, diagonalizing the tridiagonal projection reduces the regular branch to a scalar secular equation in λ , solved by 64 fixed bisection iterations.

Proposition 4 (Accepted-transaction properties). *An accepted coefficient vector c satisfies*

$$\frac{|c^\top Wc - B_0|}{\max(B_0, 1)} \leq 10^{-8}, \quad F(c) < F(\mathbf{1}). \quad (33)$$

It also has positive aggregate current-gradient and Nesterov contractions for each matrix role in every layer.

Proof. After the projected solve, the implementation recomputes the budget, the full low-rank-plus-diagonal objective, and all layer–role contractions from the candidate coefficients. The conjunction of the displayed conditions, finiteness, a regular secular branch, and the contraction predicates is the acceptance mask. A false predicate selects $c = \mathbf{1}$. $\square$

Together, these predicates certify the update budget, the quadratic loss model, and the first-order direction; the experiment section evaluates their realized next-step and endpoint behavior.

F END-TO-END UPDATE SEQUENCE

At optimizer step t , TILLER performs the following sequence.

1. Form Nesterov momentum and one grouped NS5 Muon parent for each GRAIN matrix role; form head-blocked parents for the two attention roles.
2. Update factorized row and column moments. Compute response participation and congruence on a functional refresh; otherwise restore the latest response summaries.
3. Construct the norm-preserving adaptive tangent chord for every structured role, using the local descent fallback.

4. Every eighth step, evaluate the 32 exact functional-score rows and their decay action. On other steps, create the trace-matched current-gradient rows.
5. Advance the rank-64 Robust Frequent-Directions ledger and its isotropic midpoint tail.
6. Build the 32-vector loss-Krylov basis, solve the equal-budget secular equation, and apply the global acceptance mask.
7. Multiply both GRAIN matrix directions in coordinate (ℓ, g) by the selected signed coefficient $c_{\ell g}$. Apply the scheduled update and decoupled decay once. Update attention with its response-informed chord.
8. Update the rational coefficients and all remaining parameter classes through their AdamW path.

G STATE, WORK, AND SHARDED EXECUTION

The exact method-specific floating-state inventory is

$$\text{response/window state} = 10LG + 2 + 32(LG + 1) + 4LG + 2L, \quad (34)$$

$$\text{loss ledger} = 66LG + 4, \quad (35)$$

$$\text{factorized moments} = 2L(H_r + Gd) + 6Ld. \quad (36)$$

Their sum is equation 20. Table 1 gives concrete counts used by the implementation.

Table 1: Method-specific state and largest live temporal factor. Counts exclude baseline Muon momentum and AdamW moments.

Configuration	L	G	H_r	Persistent	Live factor
12 layers, width 768	12	12	3,072	366,398	13,824
18 layers, width 1,024	18	18	4,608	976,394	31,104
Scale example, width 8,192	96	64	24,576	110,788,838	589,824

For coordinate-sharded execution, rank r holds columns $S^{(r)}$ and local vectors $h^{(r)}, w^{(r)}, r_d^{(r)}$. The identities

$$SS^\top = \sum_r S^{(r)} S^{(r)\top}, \quad \langle u, v \rangle = \sum_r \langle u^{(r)}, v^{(r)} \rangle \quad (37)$$

reduce row-Gram construction, Krylov orthogonalization, norms, budgets, and certificates to associative reductions. Tensor shards within one coordinate first reduce their scalar contractions; pipeline stages reduce the same layer-indexed summaries. The eigendecompositions remain fixed at 96 and 32, independent of C , the number of machines, and topology. A topology-aware collective may change the reduction tree without changing the mathematical result. Each shard retains its coefficient segment and applies it directly to its local parameter fragments, so no selected parameter tensor is published.

The total number of activation positions changes the activation work but not the optimizer-state formulas, score-row count, ledger rank, or solve sizes. Consequently the same state inventory applies for any N , including $N = 1,050,000$.